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ASSIGNMENT NO : BUSINESS ANALYTICS TERMINOLOGY

COURSE : DATA ANALYTICS AND DATA SCIENCE.

01 Foundational concepts of Analytics.

01 Business ~~Analytics~~ Analytics

→ It's the practice of using data, statistics and technology to analyze and make wise business decisions.

Example : Retail company analyzing the past sales on what stock was bought

02. Data Analytics

→ It is the process of collecting ~~dat~~ raw data, cleaning data and analyzing it to find patterns or trends.

Example : University checking if the students are not coping during exams

03 Business Intelligence

→ It involves using different tools to collect and analyze the data and create reports and dashboard to graphically showcase the insights.

Eg : A company that uses power BI to daily track their sales.

04. Data-Driven decision making

→ It is the practice of using data and insights to guide business decisions instead of trusting on instuition. It collects relevant data analyzing it.

Eg: Retail company analyzing the past data sales to see which product to buy, which products made more revenues.

05. Big data

→ it refers to large and more complex databases that cannot be easily managed by using traditional data processing tools.

Eg: Social media tools → it analyzes thousands of posts and Images, videos to understand user behavior and improve recommendations

06: Descriptive Analytics

→ It's analytics that only focuses on historical summarized data to understand what has happened in the business / organization

Eg: Monthly reports to show which products worked

07. Diagnostic Analytics

→ Analytics that only examines data to understand a particular trend occured.

Example: A company that invistigates why sales have dropped.

8. Predictive Analytics

→ Analytics that uses ~~hysti~~ historical data and machine learning tools to predict the Future outcomes.

Eg: Soccer apps to predict which team is likely to loose depending on their previous games played.

9. Prescriptive Analytics

→ Analytics that goes beyond prediction by recommendation of what actions that a business should take to achieve desired results

Eg: A company that uses analytics to make decisions regarding their day by day duties

10. Data Literacy

→ is the ability to be able to read and understand data effectively.

eg: Manager interpreting the dashboard data to decide which company performed best.

SECTION 2: Data, measurement and Reporting

11. Structured data

→ Data that is structured in a simplified format, usually in columns and rows.

Eg: A bank storing customer data such as account numbers, balances.

12. Unstructured data

→ Data that is not processed and does not follow a ~~structure~~ structure

eg: Social Media posts.

13. Data warehouse

→ A centralized system used to store large amounts of ~~data~~ structured data from different sources for analysis and reporting.

eg: Retail companies collecting sales from all its stores

14. Data Mining

→ The process of analyzing large datasets to discover patterns, trends and relationships

eg: Online store identies that customers buy what phones or accessories.

15. Data Visualization

→ The process of representing data in graphical formats such as charts, graphs and dashboards.

eg: An organization that uses graphs to compare monthly / quarterly sales

16. Dashboard

→ Its a visual tool used to display key data and performance metrics in real-time. It includes charts, graphs and summaries to give a quick overview.

Eg: Sales manager uses a dashboard to track daily revenues

17. Metric

→ Any measurable value used to track performance or progress in a business.

eg: A website tracks the number of visitors.

18. Key Performance Indicator (KPI)

→ A specific type of metric that is directly linked to a business goal and used to measure success.

eg: For an online store, conversion rate may be a KPI because it directly affects revenue.

19. Benchmark

→ is a standard or reference point used to compare a company's performance against industry standards or competitors

eg: A company compares its customer retention rate to the industry average.

20. Segmentation

→ is the process of dividing customers or data into groups based on shared characteristics such as age.

eg: A retail company that segments its customers into groups like students.

SECTION 3 : Customer & Sales KPI's

21. Conversion Rate

→ is the % of users or visitors who complete a desired action, such as making a purchasing.

eg: 100 people visiting a website

22. Customer Acquisition Cost (CAC)

→ its the total cost a business spends to acquire one new customer.

eg: If a company spends R10000 on marketing and gains 100 customers, the CAC is R100 per customer

23 Customer lifetime value

→ is the total profit a business expects to earn from a customer over the entire relationship

eg. A subscription service earns R200 per month from a customer who stays for 2 years resulting in a high lifetime value.

24. Churn Rate

→ It's the % of customers who stop doing business with a company over a specific period

eg: If a company starts the month with ~~Rto~~ 1000 customers and looses 50 the churn rate is 5%

25 Customer Retention Rate

→ it's the percentage of customers a business keeps over a given period

26 Net Promoter Score (NPS)

→ It measures customer loyalty by asking how likely customers are to recommend a company to others

eg: A company survey customers with a question " How likely are you to recommend us?" and calculates a score based on responses.

27. Average order value (AOV)

→ is the average amount of money a customer spends per transaction.

eg: If total revenue is R50 000 from 1000 orders, the AOV is R50

28: Sales Funnel

→ is the process that customers go through from first becoming aware of a product to making a purchase.

eg: A customer sees an ad, visits a website, compares products and then makes a purchase

Section 4: Marketing & Campaign Analytics

29. Marketing Campaign

→ A marketing campaign is planned and coordinated set of marketing activities to achieve a specific business goal

eg: A clothing brand runs a christmas special with discounts across social media and promotions.

30. Post - Campaign Analysis.

→ Post - campaign analysis is the process of performance of a marketing campaign after it ended, it measures what worked and failed.

eg: After a product launch, a company reviews sales data, web traffic

31. Incremental Revenue

→ is the additional revenue that is generated directly as a result of a marketing campaign.

eg: A business earns R200,000 more during a promotion than a normal month and that extra amount is considered incremental revenue.

32. Return on Investment (ROI)

→ Its a measure of profitability of an investment by comparing the gain to the cost.

eg: A company spends R10000 on a campaign and earns R15000 in profit resulting in positive ROI

33. Return on Ad Spend (ROAS)

→ It measures how much revenue is generated for every unit of money spent on advertising

eg: If a company spends R10000 on ads and earns R50 000 in revenu, the ROAS is 50:1

34. Click - Through Rate

→ It's the % of people who click on a link or advertisement after seeing it

eg: if 1000 people see an ad and 50 click, the CTR is 5%

35. Bounce Rate

→ is the % of website visitors who leave viewing only one page without interacting further

eg: If 100 people visit a website and 50 leave immediately, bounce rate is 50%

36. A/B Testing

→ is the method of comparing two versions of something to see which one performs better

eg. A company test two different email subjects to see which one gets more opens.

37. Lead Generation

→ is the process of attracting and capturing potential customers who may be interested in a product or service.

eg: A company offers a free eBook in exchange for a users email

38. Marketing Attribution

→ is the process of identifying which marketing channels or touchpoints contribute to a conversation or sale.

eg: A customer sees a facebook ad, clicks a google ad later and then buys

References :

1. Investopedia. (2024). Conversion Rate. Available at : https://www.investopedia.com/
2. Tableau (2024) Data visualization concepts. Available at https://www.tableau.com/
3. IBM (2024) What is Big Data? Available at https://www.ibm.com
4. Microsoft Learn (2024) Business Intelligence and analytics documentation. Available at https://learn.microsoft.com